SIMATS ENGINEERING

ASSIGNMENT - 1

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 35%

Dr.J.Velmurugan / Professor
Dept. of CSE

QUESTIONS: 5 **Total Marks: 50**

Annexure A

ASSIGNMENT - 1

Code of Conduct:

I Barath. S (192521359) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Criteria	Weightage (Marks)	Level 4 – Exemplary (Full Marks)	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Total
Conceptual Understanding	35 Marks					
Accuracy of Answers	25 Marks					
Application of Knowledge	15 Marks					
Completeness of Response	15 Marks					
Clarity & Presentation	10 Marks					
Total						

RUBRICS FOR CALCULATION

Criteria	Weightage (Marks)	Level 4 – Exemplary (Full Marks)	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	35 Marks	Demonstrates clear, complete, and in-depth understanding of OS concepts (types, structure, system calls)	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answers	25 Marks	All answers are correct, precise, and relevant	Mostly correct with minor errors	Partially correct with noticeable errors	Mostly incorrect or irrelevant
Application of Knowledge	15 Marks	Effectively applies concepts to scenarios/questions	Applies with minor mistakes	Limited or partial application	No meaningful application
Completeness of Response	15 Marks	All parts of each question are fully answered	Most parts covered with small omissions	Some parts missing	Incomplete answers
Clarity & Presentation	10 Marks	Clear, well-structured, and neatly presented answers	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Basic Terms Used in This Assignment

- **Arrival Time (AT)** – the time at which a process enters the ready queue.
- **Burst Time (BT)** – the total CPU time the process needs.
- **Completion Time (CT)** – the time at which the process finishes execution.
- **Turnaround Time (TAT)** = CT – AT, the total time from arrival to completion.
- **Waiting Time (WT)** = TAT – BT, the time spent waiting in the ready queue.
- **Gantt chart** – a bar chart of the schedule showing which process holds the CPU in each time interval (*Idle* means no process is ready).

Question 1. FCFS Scheduling

Question: Consider the set of 5 processes whose arrival time and burst time are given below. If the CPU scheduling policy is FCFS, calculate the average waiting time and average turnaround time.

Process	Arrival Time	Burst Time
P1	3	4
P2	5	3
P3	0	2
P4	5	1
P5	4	3

Answer:

Concept: First-Come, First-Served (FCFS) is the simplest CPU scheduling algorithm. It is non-preemptive: the CPU is given to processes strictly in the order in which they arrive (the ready queue is a FIFO queue), and a process keeps the CPU until it finishes. It is easy to implement and starvation-free, but short processes may wait behind a long one (the *convoy effect*), so the average waiting time can be high. If no process is in the ready queue, the CPU stays idle.

Formulas used:

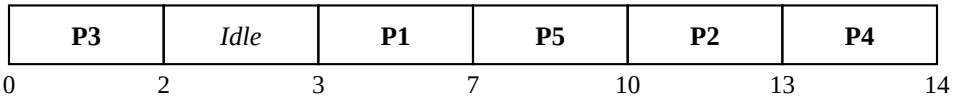
$$\textit{Turnaround Time (TAT)} = \textit{Completion Time (CT)} - \textit{Arrival Time (AT)}$$

$$\textit{Waiting Time (WT)} = \textit{TAT} - \textit{Burst Time (BT)}$$

Step-by-step scheduling:

1. **t = 0:** P3 (AT 0) is the earliest-arrived process not yet served, so it gets the CPU and runs for its full burst of 2 units: 0 → 2. CT(P3) = 2. No other process is waiting.
2. **t = 2:** the ready queue is empty (no process has arrived), so the CPU stays idle until t = 3, when P1 arrives.
3. **t = 3:** P1 (AT 3) is the earliest-arrived process not yet served, so it gets the CPU and runs for its full burst of 4 units: 3 → 7. CT(P1) = 7. No other process is waiting.
4. **t = 7:** P5 (AT 4) is the earliest-arrived process not yet served, so it gets the CPU and runs for its full burst of 3 units: 7 → 10. CT(P5) = 10. Other processes waiting in the queue: P2 (AT 5), P4 (AT 5).
5. **t = 10:** P2 (AT 5) is the earliest-arrived process not yet served, so it gets the CPU and runs for its full burst of 3 units: 10 → 13. CT(P2) = 13. Other processes waiting in the queue: P4 (AT 5).
6. **t = 13:** P4 (AT 5) is the earliest-arrived process not yet served, so it gets the CPU and runs for its full burst of 1 units: 13 → 14. CT(P4) = 14. No other process is waiting.

Gantt Chart:



Process	AT	BT	CT	TAT	WT
P1	3	4	7	4	0
P2	5	3	13	8	5
P3	0	2	2	2	0
P4	5	1	14	9	8
P5	4	3	10	6	3
Total				29	16

Calculation for each process:

Process	TAT = CT – AT	WT = TAT – BT
P1	7 – 3 = 4	4 – 4 = 0
P2	13 – 5 = 8	8 – 3 = 5
P3	2 – 0 = 2	2 – 2 = 0
P4	14 – 5 = 9	9 – 1 = 8
P5	10 – 4 = 6	6 – 3 = 3

$$AWT = \Sigma WT / n = 16/5 = 3.2$$

$$ATAT = \Sigma TAT / n = 29/5 = 5.8$$

Additional metrics: CPU busy time = 13 units out of 14, so CPU utilisation = 13/14 = 92.9%; throughput = 5/14 = 0.36 processes per unit time.

Observation: P3 is the only process at t = 0 and finishes at 2; nothing else has arrived until t = 3, so the CPU idles for 1 unit. After that, jobs are served in arrival order, and P2 and P4 (both AT = 5) wait behind P1 and P5, which is why their waiting times (5 and 8) are the largest.

Final Answer:

- Average Waiting Time = 3.2 units
- Average Turnaround Time = 5.8 units

Question 2. FCFS Scheduling

Question: Consider the set of 3 processes whose arrival time and burst time are given below. If the CPU scheduling policy is FCFS, calculate the average waiting time and average turnaround time.

Process	Arrival Time	Burst Time
P1	0	2
P2	3	1
P3	5	6

Answer:

Concept: First-Come, First-Served (FCFS) is the simplest CPU scheduling algorithm. It is non-preemptive: the CPU is given to processes strictly in the order in which they arrive (the ready queue is a FIFO queue), and a process keeps the CPU until it finishes. It is easy to implement and starvation-free, but short processes may wait behind a long one (the *convoy effect*), so the average waiting time can be high. If no process is in the ready queue, the CPU stays idle.

Formulas used:

$$Turnaround\ Time\ (TAT) = Completion\ Time\ (CT) - Arrival\ Time\ (AT)$$

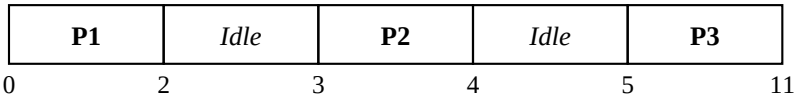
$$Waiting\ Time\ (WT) = TAT - Burst\ Time\ (BT)$$

Step-by-step scheduling:

1. **t = 0:** P1 (AT 0) is the earliest-arrived process not yet served, so it gets the CPU and runs for its full burst of 2 units: 0 → 2. CT(P1) = 2. No other process is waiting.
2. **t = 2:** the ready queue is empty (no process has arrived), so the CPU stays idle until t = 3, when P2 arrives.
3. **t = 3:** P2 (AT 3) is the earliest-arrived process not yet served, so it gets the CPU and runs for its full burst of 1 units: 3 → 4. CT(P2) = 4. No other process is waiting.

4. **t = 4:** the ready queue is empty (no process has arrived), so the CPU stays idle until t = 5, when P3 arrives.
5. **t = 5:** P3 (AT 5) is the earliest-arrived process not yet served, so it gets the CPU and runs for its full burst of 6 units: 5 → 11. CT(P3) = 11. No other process is waiting.

Gantt Chart:



Process	AT	BT	CT	TAT	WT
P1	0	2	2	2	0
P2	3	1	4	1	0
P3	5	6	11	6	0
Total				9	0

Calculation for each process:

Process	TAT = CT – AT	WT = TAT – BT
P1	2 – 0 = 2	2 – 2 = 0
P2	4 – 3 = 1	1 – 1 = 0
P3	11 – 5 = 6	6 – 6 = 0

$$AWT = \Sigma WT / n = 0/3 = 0$$

$$ATAT = \Sigma TAT / n = 9/3 = 3$$

Additional metrics: CPU busy time = 9 units out of 11, so CPU utilisation = 9/11 = 81.8%; throughput = 3/11 = 0.27 processes per unit time.
Observation: Each process arrives only after the previous one has finished, so no process ever waits: WT = 0 and TAT = BT for every process. The CPU is idle for 2 of the 11 time units (t = 2–3 and 4–5).

Final Answer:

- Average Waiting Time = 0 units
- Average Turnaround Time = 3 units

Question 3. SJF Non-Preemptive Scheduling

Question: Consider the set of 5 processes whose arrival time and burst time are given below. If the CPU scheduling policy is SJF Non-Preemptive, calculate the average waiting time and average turnaround time.

Process	Arrival Time	Burst Time
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

Answer:

Concept: Shortest Job First (non-preemptive) chooses, from the processes that have already arrived, the one with the smallest burst time and lets it run to completion. For a given set of processes it gives the minimum average waiting time among non-preemptive algorithms, but it needs the burst times in advance and long processes may starve.

Formulas used:

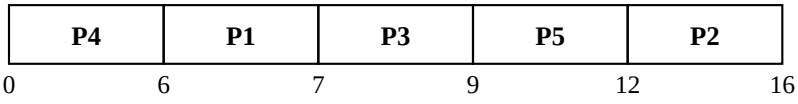
$$\textit{Turnaround Time (TAT)} = \textit{Completion Time (CT)} - \textit{Arrival Time (AT)}$$

$$\textit{Waiting Time (WT)} = \textit{TAT} - \textit{Burst Time (BT)}$$

Step-by-step scheduling:

1. **t = 0:** ready = P4 (BT 6). Smallest burst time = P4 (6), so it runs without interruption: 0 → 6. CT(P4) = 6.
2. **t = 6:** ready = P1 (BT 1), P3 (BT 2), P5 (BT 3), P2 (BT 4). Smallest burst time = P1 (1), so it runs without interruption: 6 → 7. CT(P1) = 7.
3. **t = 7:** ready = P3 (BT 2), P5 (BT 3), P2 (BT 4). Smallest burst time = P3 (2), so it runs without interruption: 7 → 9. CT(P3) = 9.
4. **t = 9:** ready = P5 (BT 3), P2 (BT 4). Smallest burst time = P5 (3), so it runs without interruption: 9 → 12. CT(P5) = 12.
5. **t = 12:** ready = P2 (BT 4). Smallest burst time = P2 (4), so it runs without interruption: 12 → 16. CT(P2) = 16.

Gantt Chart:



Process	AT	BT	CT	TAT	WT
P1	3	1	7	4	3
P2	1	4	16	15	11
P3	4	2	9	5	3
P4	0	6	6	6	0
P5	2	3	12	10	7
Total				40	24

Calculation for each process:

Process	TAT = CT – AT	WT = TAT – BT
P1	7 – 3 = 4	4 – 1 = 3
P2	16 – 1 = 15	15 – 4 = 11
P3	9 – 4 = 5	5 – 2 = 3
P4	6 – 0 = 6	6 – 6 = 0
P5	12 – 2 = 10	10 – 3 = 7

$$AWT = \Sigma WT / n = 24/5 = 4.8$$

$$ATAT = \Sigma TAT / n = 40/5 = 8$$

Additional metrics: CPU busy time = 16 units out of 16, so CPU utilisation = 16/16 = 100.0%; throughput = 5/16 = 0.31 processes per unit time.
Observation: P4 is the only process at t = 0, so it starts first and cannot be interrupted (non-preemptive). At t = 6 the shortest ready job, P1, goes next, then P3 and P5, and the longest ready job, P2, runs last, which gives it the largest waiting time (11).

Final Answer:

- Average Waiting Time = 4.8 units
- Average Turnaround Time = 8 units

Question 4. SJF Preemptive Scheduling

Question: Consider the set of 5 processes whose arrival time and burst time are given below. If the CPU scheduling policy is SJF Preemptive, calculate the average waiting time and average turnaround time.

Process	Arrival Time	Burst Time
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

Answer:

Concept: Preemptive SJF, also called Shortest Remaining Time First (SRTF), re-examines the ready queue every time a process arrives or finishes. If a newly arrived process has a burst time smaller than the *remaining* time of the running process, the running process is pre-empted. It gives the smallest possible average waiting time but causes more context switches and can starve long processes.

Formulas used:

$$Turnaround\ Time\ (TAT) = Completion\ Time\ (CT) - Arrival\ Time\ (AT)$$

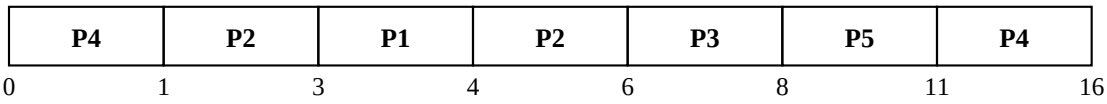
$$Waiting\ Time\ (WT) = TAT - Burst\ Time\ (BT)$$

Step-by-step scheduling:

1. **t = 0:** ready = P4 (rem 6). Shortest remaining time = P4 (6), runs 0 → 1. At t = 1, P2 arrives with a shorter remaining time than P4's 5, so P4 is pre-empted with 5 unit(s) left.

2. **t = 1:** ready = P2 (rem 4), P4 (rem 5). Shortest remaining time = P2 (4), runs 1 → 3. At t = 3, P1 arrives with a shorter remaining time than P2's 2, so P2 is pre-empted with 2 unit(s) left.
3. **t = 3:** ready = P1 (rem 1), P2 (rem 2), P5 (rem 3), P4 (rem 5). Shortest remaining time = P1 (1), runs 3 → 4. P1 completes at t = 4. CT(P1) = 4.
4. **t = 4:** ready = P2 (rem 2), P3 (rem 2), P5 (rem 3), P4 (rem 5). Shortest remaining time = P2 (2), runs 4 → 6. P2 completes at t = 6. CT(P2) = 6.
5. **t = 6:** ready = P3 (rem 2), P5 (rem 3), P4 (rem 5). Shortest remaining time = P3 (2), runs 6 → 8. P3 completes at t = 8. CT(P3) = 8.
6. **t = 8:** ready = P5 (rem 3), P4 (rem 5). Shortest remaining time = P5 (3), runs 8 → 11. P5 completes at t = 11. CT(P5) = 11.
7. **t = 11:** ready = P4 (rem 5). Shortest remaining time = P4 (5), runs 11 → 16. P4 completes at t = 16. CT(P4) = 16.

Gantt Chart:



Process	AT	BT	CT	TAT	WT
P1	3	1	4	1	0
P2	1	4	6	5	1
P3	4	2	8	4	2
P4	0	6	16	16	10
P5	2	3	11	9	6
Total				35	19

Calculation for each process:

Process	TAT = CT – AT	WT = TAT – BT
P1	4 – 3 = 1	1 – 1 = 0
P2	6 – 1 = 5	5 – 4 = 1
P3	8 – 4 = 4	4 – 2 = 2
P4	16 – 0 = 16	16 – 6 = 10
P5	11 – 2 = 9	9 – 3 = 6

$$AWT = \Sigma WT / n = 19/5 = 3.8$$

$$ATAT = \Sigma TAT / n = 35/5 = 7$$

Additional metrics: CPU busy time = 16 units out of 16, so CPU utilisation = 16/16 = 100.0%; throughput = 5/16 = 0.31 processes per unit time.

Observation: Using the same data as Question 3, pre-emption lowers AWT from 4.8 to 3.8 and ATAT from 8 to 7. The long job P4 is pre-empted at t = 1 and resumes only at t = 11, so it pays the highest price (WT = 10), whereas the short jobs P1 and P2 hardly wait.

Final Answer:

- Average Waiting Time = 3.8 units
- Average Turnaround Time = 7 units

Question 5. Round Robin Scheduling

Question: Consider the set of 5 processes whose arrival time and burst time are given below. If the CPU scheduling policy is Round Robin with time quantum = 2 units, calculate the average waiting time and average turnaround time.

Process	Arrival Time	Burst Time
P1	0	5
P2	1	3
P3	2	1
P4	3	2
P5	4	3

Answer:

Concept: Round Robin (RR) is a preemptive algorithm designed for time-sharing. Each process receives the CPU for at most one time quantum; if it has not finished, it is pre-empted and placed at the end of the ready queue. When a process arrives at the same moment another one's quantum expires, the new arrival is queued first. RR is fair and starvation-free, but its waiting time depends strongly on the quantum.

Formulas used:

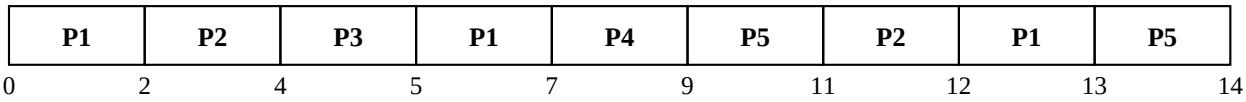
$$\textit{Turnaround Time (TAT)} = \textit{Completion Time (CT)} - \textit{Arrival Time (AT)}$$

$$\textit{Waiting Time (WT)} = \textit{TAT} - \textit{Burst Time (BT)}$$

Step-by-step scheduling:

1. **t = 0:** ready queue = [P1]. P1 runs for min(2, 5) = 2 unit(s): 0 → 2. P1 has 3 unit(s) left and goes to the back of the queue.
2. **t = 2:** ready queue = [P2, P3, P1]. P2 runs for min(2, 3) = 2 unit(s): 2 → 4. P2 has 1 unit(s) left and goes to the back of the queue.
3. **t = 4:** ready queue = [P3, P1, P4, P5, P2]. P3 runs for min(2, 1) = 1 unit(s): 4 → 5. P3 finishes. CT(P3) = 5.
4. **t = 5:** ready queue = [P1, P4, P5, P2]. P1 runs for min(2, 3) = 2 unit(s): 5 → 7. P1 has 1 unit(s) left and goes to the back of the queue.
5. **t = 7:** ready queue = [P4, P5, P2, P1]. P4 runs for min(2, 2) = 2 unit(s): 7 → 9. P4 finishes. CT(P4) = 9.
6. **t = 9:** ready queue = [P5, P2, P1]. P5 runs for min(2, 3) = 2 unit(s): 9 → 11. P5 has 1 unit(s) left and goes to the back of the queue.
7. **t = 11:** ready queue = [P2, P1, P5]. P2 runs for min(2, 1) = 1 unit(s): 11 → 12. P2 finishes. CT(P2) = 12.
8. **t = 12:** ready queue = [P1, P5]. P1 runs for min(2, 1) = 1 unit(s): 12 → 13. P1 finishes. CT(P1) = 13.
9. **t = 13:** ready queue = [P5]. P5 runs for min(2, 1) = 1 unit(s): 13 → 14. P5 finishes. CT(P5) = 14.

Gantt Chart:



Process	AT	BT	CT	TAT	WT
P1	0	5	13	13	8
P2	1	3	12	11	8
P3	2	1	5	3	2
P4	3	2	9	6	4
P5	4	3	14	10	7
Total				43	29

Calculation for each process:

Process	TAT = CT – AT	WT = TAT – BT
P1	13 – 0 = 13	13 – 5 = 8
P2	12 – 1 = 11	11 – 3 = 8
P3	5 – 2 = 3	3 – 1 = 2
P4	9 – 3 = 6	6 – 2 = 4
P5	14 – 4 = 10	10 – 3 = 7

$$AWT = \Sigma WT / n = 29/5 = 5.8$$

$$ATAT = \Sigma TAT / n = 43/5 = 8.6$$

Additional metrics: CPU busy time = 14 units out of 14, so CPU utilisation = 14/14 = 100.0%; throughput = 5/14 = 0.36 processes per unit time.

Observation: With a quantum of 2, every process gets its first turn quickly (good response time), but long jobs such as P1 are repeatedly pre-empted and finish late (CT = 13), so the waiting times are relatively high.

Final Answer:

- Average Waiting Time = 5.8 units
- Average Turnaround Time = 8.6 units

Final Answer Summary

Q.No	Algorithm	Average Waiting Time	Average Turnaround Time
1	FCFS	3.2	5.8
2	FCFS	0	3
3	SJF Non-Preemptive	4.8	8
4	SJF Preemptive	3.8	7
5	Round Robin	5.8	8.6

Note: All values are in time units.

Comparison and Conclusion

- Questions 3 and 4 use identical data. SJF Non-Preemptive gives AWT = 4.8, whereas SJF Preemptive (SRTF) gives AWT = 3.8, so pre-emption reduced the average waiting time by 1 unit and the average turnaround time from 8 to 7.
- FCFS (Questions 1 and 2) is simple, but its waiting time depends on the arrival order. In Question 2 every process arrives after the previous one finishes, so nobody waits; in Question 1 a burst of arrivals queues up behind earlier jobs.
- Round Robin (Question 5) is the fairest, because every process gets the CPU regularly, but it does not minimise waiting time.
- In general: SRTF/SJF minimise average waiting time but can starve long processes; FCFS is fair in arrival order but can suffer the convoy effect; Round Robin is best suited to time-sharing systems.